

Photochemistry

Introduction

The Photochemistry course is concerned with the interaction of visible and ultraviolet light with molecules, an important aspect of modern chemistry which is relevant to biology (*e.g.* photosynthesis, vision), lasers, organic synthesis, reaction kinetics and atmospheric science (*e.g.* the ozone hole). Some familiarity with concepts such as Hund's rules, the Franck-Condon principle, basic reaction kinetics and the steady-state approximation is expected.

Suggested Reading

1. Atkins, P. W., *Physical Chemistry*, any edition, OUP.
2. Wayne, C. E., and Wayne R. P., *Photochemistry*, OUP Primer 39, 1996.
3. Wayne, R. P., *Principles and Applications of Photochemistry*, 2nd ed., OUP, 1988.
4. Gilbert, A., and Baggott, J., *Essentials of Molecular Photochemistry*, Blackwell, 1991.
5. Albani, J. R., *Principles and Applications of Fluorescence Spectroscopy*, Blackwell, 2007.
6. Turro, N. J., *Modern Molecular Photochemistry*, University Science Books, 1991.
7. Suppan, P., *Chemistry and Light*, RSC, 1994.
8. Lakowicz, J. R., *Principles of Fluorescence Spectroscopy*, 2nd ed., Springer, 1999.

Notation and Common Pitfalls

Units and accuracy

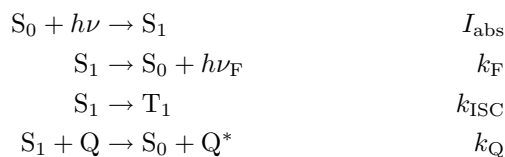
As ever, give your answers to an appropriate number of significant figures, with their correct units, and label the axes of your graphs clearly.

Frequency and wavelength

The frequency, ν , of a photon involved in a transition between two levels in a molecule is proportional to their energy separation: $h\nu = \Delta E = E_2 - E_1$. The wavelength is therefore *inversely* related to ΔE : $\lambda = hc/\Delta E$.

Rates and rate constants of photochemical processes

A simple sequence of photochemical processes may be written



Note that k_F , k_{ISC} and k_Q are *rate constants* of first, first, and second order respectively (remember this when you give their units) but I_{abs} , the intensity of the absorbed radiation, is a *rate*. Therefore, if $[S_1]$ is in steady state its rate of formation ($= I_{abs}$) is equal to its rate of destruction:

$$I_{abs} = k_F[S_1] + k_{ISC}[S_1] + k_Q[S_1][Q].$$

Radiative lifetime and observed lifetime

The *radiative lifetime* of a species, such as S_1 above, is its lifetime *in the absence of radiationless transitions*: $\tau_{0,F} = 1/k_F$. This is, in general, different from the *observed lifetime* (or ‘*apparent lifetime*’, or just the ‘*lifetime*’) which is the lifetime of the species *due to all the processes contributing to its removal*. For example, imagine a population of $[S_1]_0$ is created by a pulse of light and is allowed to decay away through the above three processes (with no subsequent absorption of light). The $[S_1]$ concentration would change according to:

$$-\frac{d[S_1]}{dt} = k_F[S_1] + k_{ISC}[S_1] + k_Q[S_1][Q].$$

This is a first-order ordinary differential equation with the solution

$$\begin{aligned} [S_1] &= [S_1]_0 \exp [-(k_F + k_{ISC} + k_Q[Q])t] \\ &= [S_1]_0 e^{-t/\tau}, \end{aligned}$$

and so the observed lifetime is $\tau = 1/(k_F + k_{ISC} + k_Q[Q])$. The fluorescence intensity at a given instant in time is proportional to the S_1 concentration: $I_F = k_F[S_1]$ and so τ is also the *observed lifetime of fluorescence* (often referred to as the *fluorescence lifetime* and not to be confused with $\tau_{0,F}$ which is only equal to τ in the absence of intersystem crossing and quenching). Unfortunately, there is a lot of inconsistent notation in books, lecture notes and past exam papers surrounding this issue so be very careful to determine which τ you are using.

Quantum yield

The quantum yield for a process may be defined as the number of times a specified photochemical event occurs divided by the number of photons absorbed. Equivalently, it is the rate of the specified photochemical process divided by the sum of the rates of all the processes which remove the initially excited species. For example,

$$\begin{aligned} \Phi_F &= \frac{k_F[S_1]}{k_F[S_1] + k_{ISC}[S_1] + k_Q[S_1][Q]} \\ &= \frac{k_F}{k_F + k_{ISC} + k_Q[Q]}. \end{aligned}$$

Similarly, the quantum yield for triplet formation via intersystem crossing is

$$\Phi_T = \frac{k_{ISC}}{k_F + k_{ISC} + k_Q[Q]}.$$

Now, suppose the triplet state undergoes either phosphorescence or intersystem crossing to the ground state:



What is the quantum yield for phosphorescence? Not

$$\Phi_{\text{P}} = \frac{k_{\text{P}}}{k_{\text{P}} + k_{\text{ISC}}} \quad (\text{wrong})$$

because the quantum yield refers to the photons which created the initial excited electronic state (S_1): it is not equal to the fraction of $[\text{T}_1]$ that phosphoresces but the fraction of $[\text{S}_1]$ that undergoes intersystem crossing to T_1 which then phosphoresces:

$$\Phi_{\text{P}} = \Phi_{\text{T}} \frac{k_{\text{P}}}{k_{\text{P}} + k_{\text{ISC}}}.$$

Note that this means that the equation at the bottom of p.847 of Atkins 8th edition is incorrect, *i.e.* in general it is not true that $\Phi_{\text{F}} + \Phi_{\text{T}} + \Phi_{\text{P}} + \dots = 1$.

The Beer-Lambert law

The Beer-Lambert Law may be written as

$$I = I_0 e^{-\alpha l} \quad \text{or} \quad I = I_0 10^{-\epsilon l}$$

where α and ϵ are the *Napierian* and *decadic* absorption coefficients, related by $\alpha = \epsilon \ln 10$. Note that $I = I_{\text{trans}}$ here is the *transmitted* intensity. The absorbed intensity, $I_{\text{abs}} = I_0 - I_{\text{trans}}$.

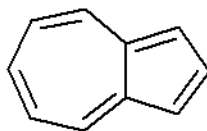
Notes

Make your own notes for use in revision on the following topics. Please don't hand them in: I'm sure they're lovely.

- The Jablonski diagram, illustrating the following processes: fluorescence, internal conversion (IC), intersystem crossing (ISC), phosphorescence, vibrational relaxation (VR).
- Quantum yield; radiative lifetime and observed lifetime.
- The Franck-Condon principle; Kasha rule; mirror-image absorption and fluorescence spectra; the mechanism of intersystem crossing; El Sayed rules.
- Photochemical kinetics; the Stern-Volmer plot.
- Selection rules for electronic transitions: deducing whether a transition is allowed or forbidden from the symmetries of the electronic wavefunctions involved in the molecular point group; vibronic coupling.
- The Beer-Lambert law; its applications and breakdown.
- Photoionization and photoelectron spectroscopy.
- Intermolecular energy transfer; Förster cycles.

Questions

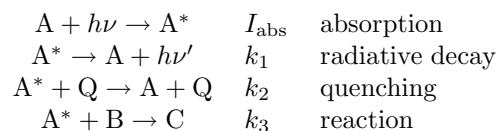
- Draw a schematic diagram (a *Jablonski* diagram) to show the relative energies of the ground state and excited electronic states of a polyatomic molecule and the possible radiative and non-radiative transitions between them.
 - Show that the quantum yield, Φ_F , the radiative lifetime, $\tau_{0,F}$, and the observed lifetime, τ , in fluorescence are related by the equation $\Phi_F = \tau/\tau_{0,F}$. Here $\tau = 1/(k_F + k_{NR})$, where k_F is the fluorescence rate constant and k_{NR} is the sum of the rate constants for non-radiative processes depopulating the excited state. Quenching may be assumed to be absent.
 - For a certain aromatic hydrocarbon in a glassy matrix at low temperature, excited to the S_1 state, the quantum yield of fluorescence is 0.34, the quantum yield for triplet formation is 0.66, and the quantum yield of phosphorescence is 0.024. The measured lifetimes of fluorescence and phosphorescence are 118 ns and 2.6 s respectively.
 - Determine the rate constant for intersystem crossing from S_1 to T_1 .
 - Determine the rate constant for intersystem crossing from T_1 to S_0 .
 - Comment on the difference between the two lifetimes and the two intersystem crossing rate constants.
- Give explanations for the following observations, as completely as possible, *giving diagrams where appropriate*.
 - Fluorescence is generally observed only from the first excited singlet state, S_1 , but for azulene (**1**) fluorescence is observed from the second excited singlet state, S_2 , as well. The energies of the singlet states in azulene are $E(S_1 - S_0) = 14200 \text{ cm}^{-1}$ and $E(S_2 - S_0) = 28300 \text{ cm}^{-1}$.



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- The intersystem crossing rates in halobenzenes increase with the atomic number of the halogen.
- Solid matrix studies show that naphthalene ($C_{10}H_8$) has a shorter phosphorescence lifetime than perdeuteronaphthalene ($C_{10}D_8$).
- The rate constant for intersystem crossing from S_1 to T_1 in propanone is $5 \times 10^8 \text{ s}^{-1}$, but in benzophenone it is roughly 10^{11} s^{-1} .
- The fluorescence spectrum of large aromatic molecules in solution is often a 'mirror image' of its absorption spectrum. The features corresponding to the 0-0 vibrational transition in each spectrum are at almost (but not quite) the same wavelength. Biphenyl does not display this behaviour.

3. The photoreduction of benzophenone (A) may be represented by the reaction scheme shown below:

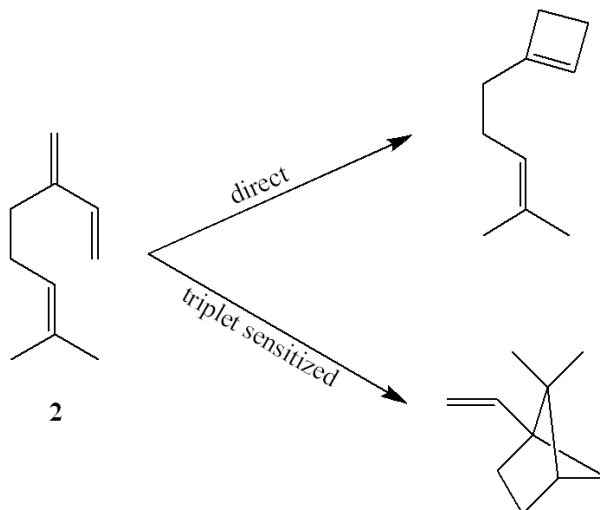


Derive an expression for the quantum yield of C formation, Φ , for the different concentrations of Q and B given below. Show that they are consistent with the scheme above and obtain values for the ratios k_1/k_3 and k_2/k_3 .

[Q] = 0				
[B] /M	1.0	0.2	0.07	0.04
Φ	0.97	0.78	0.52	0.4
[Q] = 10^{-3} M				
[B] /M	0.25	0.09	0.04	0.02
Φ	0.5	0.25	0.125	0.07

Assuming that k_2 has the typical diffusion-controlled value of $5 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$, calculate k_1 and k_3 . The radiative lifetime of the A^* singlet state is estimated to be $5 \times 10^{-5} \text{ s}$, but its fluorescence quantum yield is 10^{-4} . Comment on these values and the value you have obtained for k_1 .

4. What is meant by *triplet sensitization* in the context of photoreactivity? Explain why different products are observed on direct and triplet-sensitized irradiation of myrcene (**2**) and suggest mechanisms for their production.



5. The radiative lifetime of the first excited singlet state in benzene, of $^1B_{2u}$ symmetry, is 414 ns, but its measured lifetime in solution is much shorter. For anthracene, the radiative and measured lifetimes of the first excited singlet state (also labelled $^1B_{2u}$) are both about 10 ns. Show that these observations are consistent with the formal selection rules derived from group theory considerations.
6. (a) Ultraviolet radiation photolyses O_3 to O_2 and O. Determine the rate at which ozone is consumed (per unit area) by 305 nm radiation in a layer of the stratosphere of thickness 1 km. The quantum yield for the photodissociation is 0.94 at 220 K, the concentration of O_3 is about $8 \times 10^{-9} \text{ mol dm}^{-3}$, the molar decadic absorption coefficient is $260 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$, and the intensity of radiation at this wavelength is about $10^{14} \text{ photons cm}^{-2} \text{ s}^{-1}$.
- (b) A solution of $K_3[Fe(C_2O_4)_3]$ of concentration $0.006 \text{ mol dm}^{-3}$ is contained in a cell of thickness 1 mm and is irradiated with light of wavelength 400 nm. The molar decadic absorption coefficient at this wavelength is $258 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$. What fraction of the incident light intensity is absorbed?

The Fe(III) is reduced to Fe(II) on irradiation and the quantum yield for Fe(II) formation at $\lambda = 400 \text{ nm}$ is 1.13. If the solution is irradiated for 20 minutes with light of $\lambda = 400 \text{ nm}$, it is found that $3.4 \times 10^{-6} \text{ mol}$ of Fe(II) are formed. What is the intensity of the light incident on the cell (measured in quanta s^{-1})?

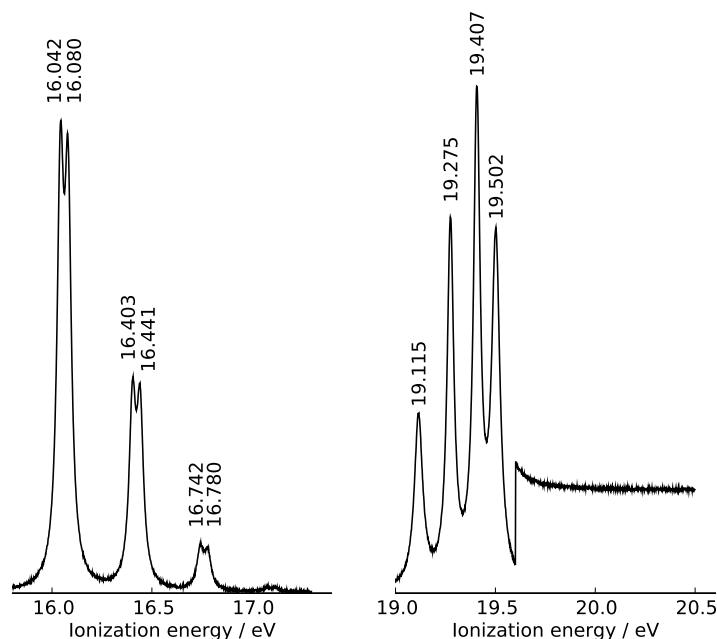
Explain how this problem is connected to the idea of a *chemical actinometer*.

- (c) The molar decadic absorption coefficient, ϵ , for a compound is $3 \times 10^4 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$ at 313 nm where the compound undergoes a photochemical bleaching reaction. The product does not absorb at 313 nm.

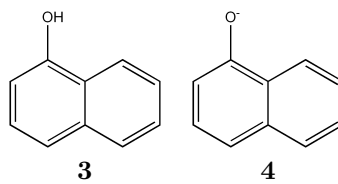
A solution of the compound, at concentration $c = 1.5 \times 10^{-3} \text{ M}$, in a 1 cm^3 cell of 1 cm^2 cross section was placed in the path of a light beam of 0.01 mW cm^{-2} at 313 nm. What fraction of the incident intensity is absorbed?

The *absorbance* (this quantity is equal to ϵcl where l is the path length of the light) decreased by 7% of the original value in two hours. Obtain the quantum yield of the bleaching reaction.

7. (a) The photoelectron spectrum of HF excited by 40.81 eV photons, shown below, consists of two band systems. Sketch a molecular orbital diagram for HF and identify the orbitals from which the electrons are ejected and the states of HF^+ formed by ionization.
- (b) Explain the form of the lower ionization band system and estimate the vibrational constants ω_e and $\omega_e x_e$ for the corresponding state of HF^+ .
- (c) Estimate the dissociation energy of HF^+ in the excited electronic state from the higher ionization energy band system and comment on the appearance of this band in the light of your answer.



8. The maximum in the absorption spectrum of 1-naphthol (**3**) is shifted by $+3300 \text{ cm}^{-1}$ with respect to the maximum in the absorption spectrum of its conjugate base (**4**). Estimate the difference in pK_a values for the S_0 and S_1 states of 1-naphthol at 300 K.



Answers to Selected Questions

1. (c)(i) $k_{\text{ISC}} = 5.60 \times 10^6 \text{ s}^{-1}$, (ii) $k'_{\text{ISC}} = 0.371 \text{ s}^{-1}$
3. $k_1 = 1.51 \times 10^6 \text{ s}^{-1}$, $k_3 = 2.44 \times 10^7 \text{ M}^{-1} \text{ s}^{-1}$
6. (a) $3.58 \times 10^{13} \text{ molec cm}^{-2} \text{ s}^{-1}$, or $3.58 \times 10^8 \text{ molec cm}^{-3} \text{ s}^{-1}$ for the change of concentration within the entire layer; (b) 0.3, $5.03 \times 10^{15} \text{ quanta s}^{-1}$, (c) $\Phi = 0.55$
7. (b) $\omega_e = 0.383 \text{ eV}$, $\omega_e x_e = 0.011 \text{ eV}$, (c) $D_0 \approx 0.485 \text{ eV}$